



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT		
S. No.	University ID	Name
1	2420030543	Pardha Pranav
2	2420030190	Sai Sri Harsha
3	2420030220	Sandeep
4	2420030216	Mohit Nandan Reddy
Name of the Guide		RAJKUMAR PATIL

Abstract

The **Predictive Maintenance Management System** is a software-based solution designed to predict potential equipment failures before they occur and help organizations schedule maintenance activities efficiently. Traditional maintenance approaches often depend on fixed schedules or repairs after equipment failure, which can result in unexpected downtime, increased maintenance costs, and reduced equipment efficiency. This project addresses these limitations by using historical machine data and machine learning techniques to identify equipment that may require maintenance.

The system collects and processes parameters such as temperature, rotational speed, torque, tool wear, operating hours, and previous failure records. Data preprocessing techniques are applied to clean and transform the dataset before it is provided to a machine learning model. A **Random Forest classification algorithm** is used to predict the risk of machine failure. The prediction results are presented through an interactive dashboard that displays machine health, failure probability, maintenance status, and alerts for high-risk equipment.

The system also provides features for machine management, maintenance scheduling, maintenance history, and data visualization. The project follows an **adaptive and iterative software development approach**, allowing features to be developed, tested, evaluated, and improved based on feedback during different development iterations. By combining software engineering, data analytics, and machine learning, the proposed system can help reduce unexpected equipment failures, minimize downtime, improve maintenance planning, and support data-driven decision-making.

Signature of the Guide

HOD